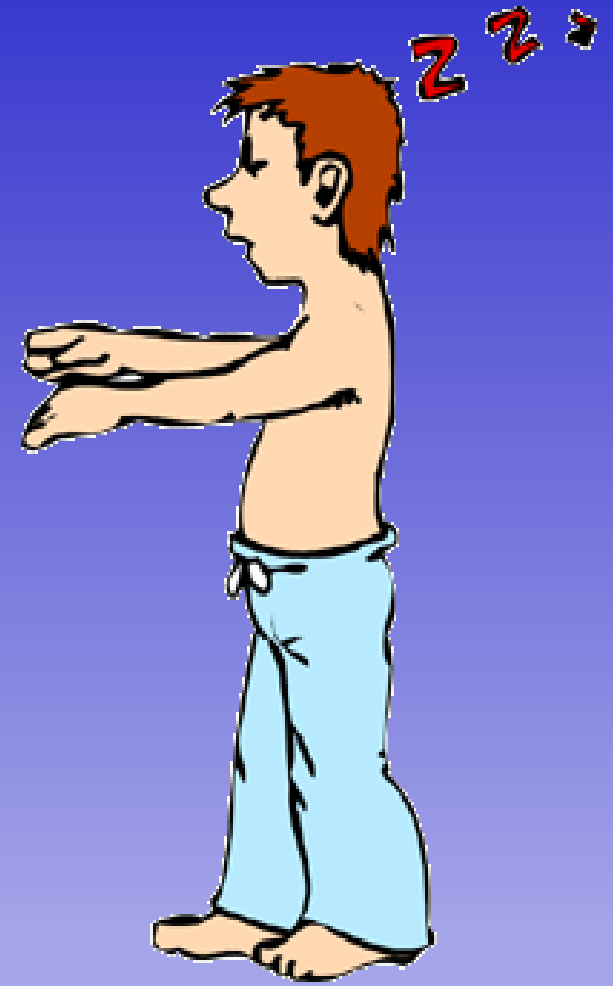


Parasomnias and Sleep Forensics

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M E D I C I N E

Parasomnias

“...are undesirable events or experiences that occur during entry into sleep, within sleep, or during arousals from sleep. These events are manifestations of central nervous system activation transmitted into skeletal muscle and autonomic nervous system channels, often with experiential concomitants. Parasomnias encompass abnormal sleep related movements, behaviors, emotions, perceptions, dreaming, autonomic nervous system functioning. Furthermore, “basic drive states” can emerge in pathologic forms ...as seen with sleep related aggression and locomotion, SRED, and abnormal sleep related sexual behaviors. Parasomnias are clinical disorders because of the resulting injuries, sleep disruption, adverse health effects, and untoward psychosocial effects.”

Parasomnias ICSD 3

- **NREM-Related Parasomnias**
 1. Disorders of Arousal (From NREM Sleep)
 - a) Confusional Arousals
 - b) Sleepwalking
 - c) Sleep Terrors
 2. Sleep Related Eating Disorder

Parasomnias ICSD 3

- **REM-Related Parasomnias**
 1. REM Sleep Behavior Disorder
 2. Recurrent Isolated Sleep Paralysis
 3. Nightmare Disorder

Parasomnias ICSD 3

- **Other Parasomnias**

1. Exploding Head Syndrome
2. Sleep Related Hallucinations
3. Sleep Enuresis
4. Parasomnia Due to a Medical Disorder
5. Parasomnia Due to a Medication or Substance
6. Parasomnia, Unspecified

Parasomnias

- **Isolated Symptoms and Normal Variants**
Sleep Talking

Nightmares



Dream Rehearsal



Who made Max king of the Wild Things?

Dream Rehearsal



Old Hag Syndrome

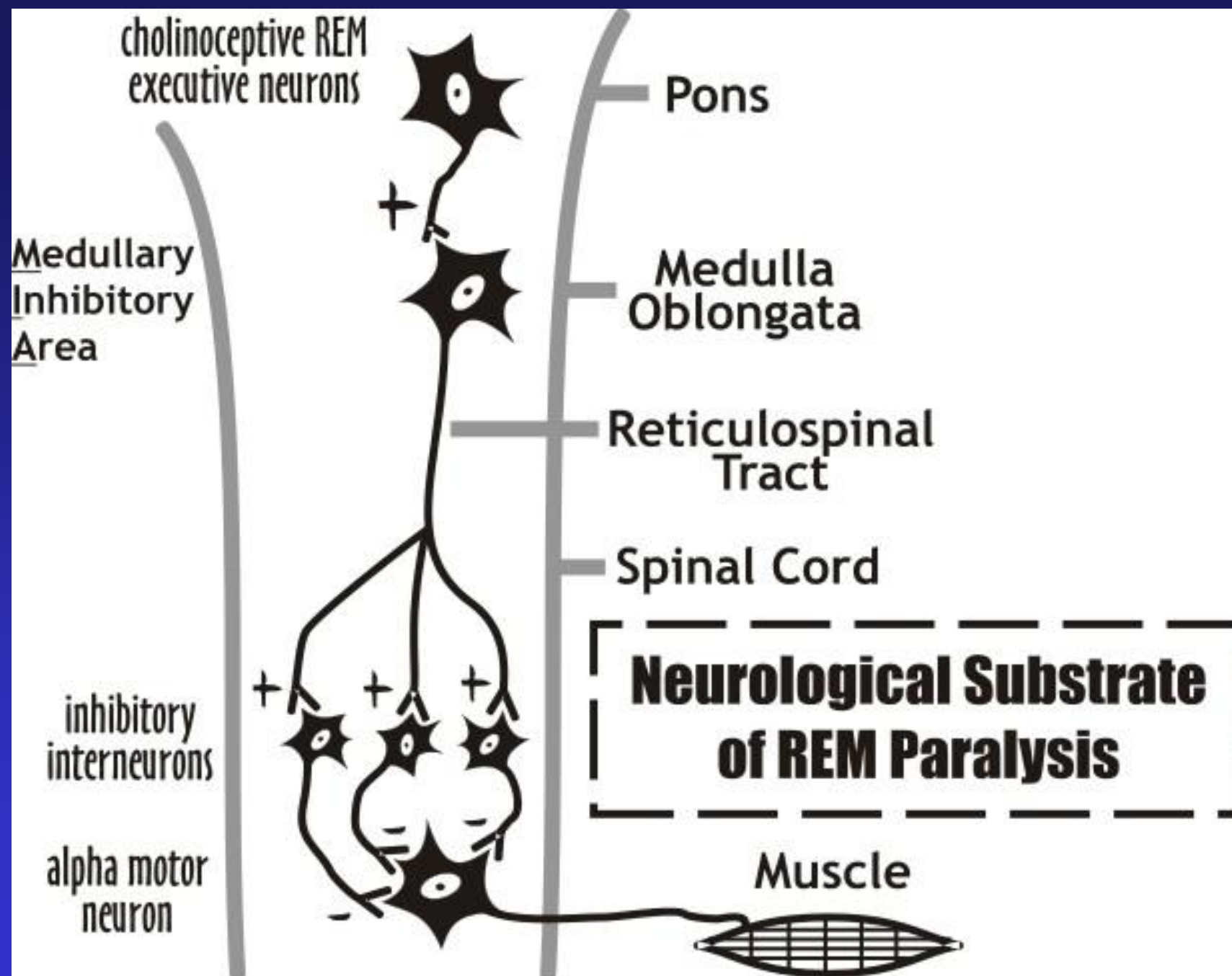


Old Hag Syndrome

- Sleep paralysis with hypnagogic and hypnopompic hallucinations has been singled out as a particularly likely source of beliefs concerning supernatural phenomenon
- Patients may report feeling a “presence” that can be perceived as threatening or evil. An intense sense of dread and terror is very common

SLEEP PARALYSIS

- Transient inability to perform voluntary movements upon awakening. Episodes can vary in duration from seconds to several minutes
- Episodes are usually very distressing, especially when accompanied by hypnagogic or hypnopompic hallucinations.
- Patients may become less frightened over time with repeated episodes



Ghost on Chest

“This is not a dream...When I awake from my dream and try to get out the bed I can't get up. I can't move I can't roll off the bed it feels like something is sitting on my chest. This is very real when I finally get out the bed it's like I'm out of breath, my heartbeat is abnormal.

This has happened to me on more than one occasion. I feel like I'm being told by a ghost to kill myself ... I would never do such a thing but this is the vibe that I get from this. I have to say that it not only happens at my home but at friends houses that I have stayed at. So I know it's not the house,...

I have spoken about this to a number of people and they said they have had the same thing happen to them. I've tried to compare any similarities in the people I've asked and all I could get is that we are all around the same age, which is 19-21. If this happened to more than one person is this what the people who commit suicide experience before the they commit such a thing.

Does a ghostly spirit visit us every night and some of the spirits stay a tad bit longer because they like your **sole**. I don't know what to think. What do you think of this?”

<http://www.yourghoststories.com/real-ghost-story.php?story=6939>

Sleep Paralysis Treatment

- Sleep paralysis episodes may occur more commonly in patients who are sleep deprived, stressed, or have very irregular sleep schedules. Sleep paralysis is relatively common in the general population and more frequent in students and psychiatric patients. Attacks may be avoided by maintaining a regular sleep schedule and decreasing stress.
- Medications are not usually indicated for isolated sleep paralysis. Medications that improve cataplexy may also reduce sleep paralysis episodes

Confusional Arousals

- Consist of confusion during and following arousals from sleep, usually from deep sleep in the first part of the night.
- Disoriented in time and space, slow of speech and mentation, and responds poorly and slowly to command or questioning. There is major memory impairment, both retrograde and anterograde in type. Behavior may be very inappropriate, such as picking up a lamp to talk when the sleeper believed the telephone had rung. The confusional behavior may last from several minutes to several hours. Confusional arousals can be precipitated by forced awakenings, mainly in the first third of the night.

Sleepwalking

- Consists of a series of complex behaviors that are initiated during slow-wave sleep and result in walking during sleep.
- Episodes can range from simple sitting up in bed to walking and even to apparent frantic attempts to "escape." The patient may be difficult to awaken but, when awakened, often is mentally confused. The patient is usually amnesic for the episode's events. Sleepwalking is most often evident during the first third of the night or during other times of increased slow-wave sleep, such as after sleep deprivation. The motor activity may terminate spontaneously, or the sleepwalker may return to bed, lie down, and continue to sleep without reaching alertness at any point.

sleepwalking

Sleep Terrors

- Characterized by a sudden arousal from slow-wave sleep with a piercing scream or cry, accompanied by autonomic and behavioral manifestations of intense fear.
- Severe autonomic discharge, with tachycardia, tachypnea, flushing of the skin, diaphoresis, mydriasis, decreased skin resistance, and increased muscle tone. The patient usually sits up in bed, is unresponsive to external stimuli, and, if awakened, is confused and disoriented. Amnesia occurs, although sometimes there are reports of fragments or very brief vivid dream images. The episode may be accompanied by incoherent vocalizations or micturition.

Sleep Terrors

pavor
nocturnus

Night Terror



Sleep Terrors vs Nightmares

- The nightmare sufferer usually remembers the dream content in vivid detail.
- Nightmares occur during the last third of the night, as opposed to sleep terrors, which occur at the beginning. Nightmares usually do not involve major motor activity.
- There is considerably less anxiety, vocalization, and autonomic discharge during a nightmare than during a sleep terror.
- When awakened during a nightmare, the patient exhibits good intellectual function, whereas the patient with sleep terror is usually confused.
- Nightmares occur out of REM sleep, as opposed to out of slow-wave sleep for sleep terrors.

Arousal Parasomnia Treatment

Safety

Safety

Safety

Avoid sleep deprivation

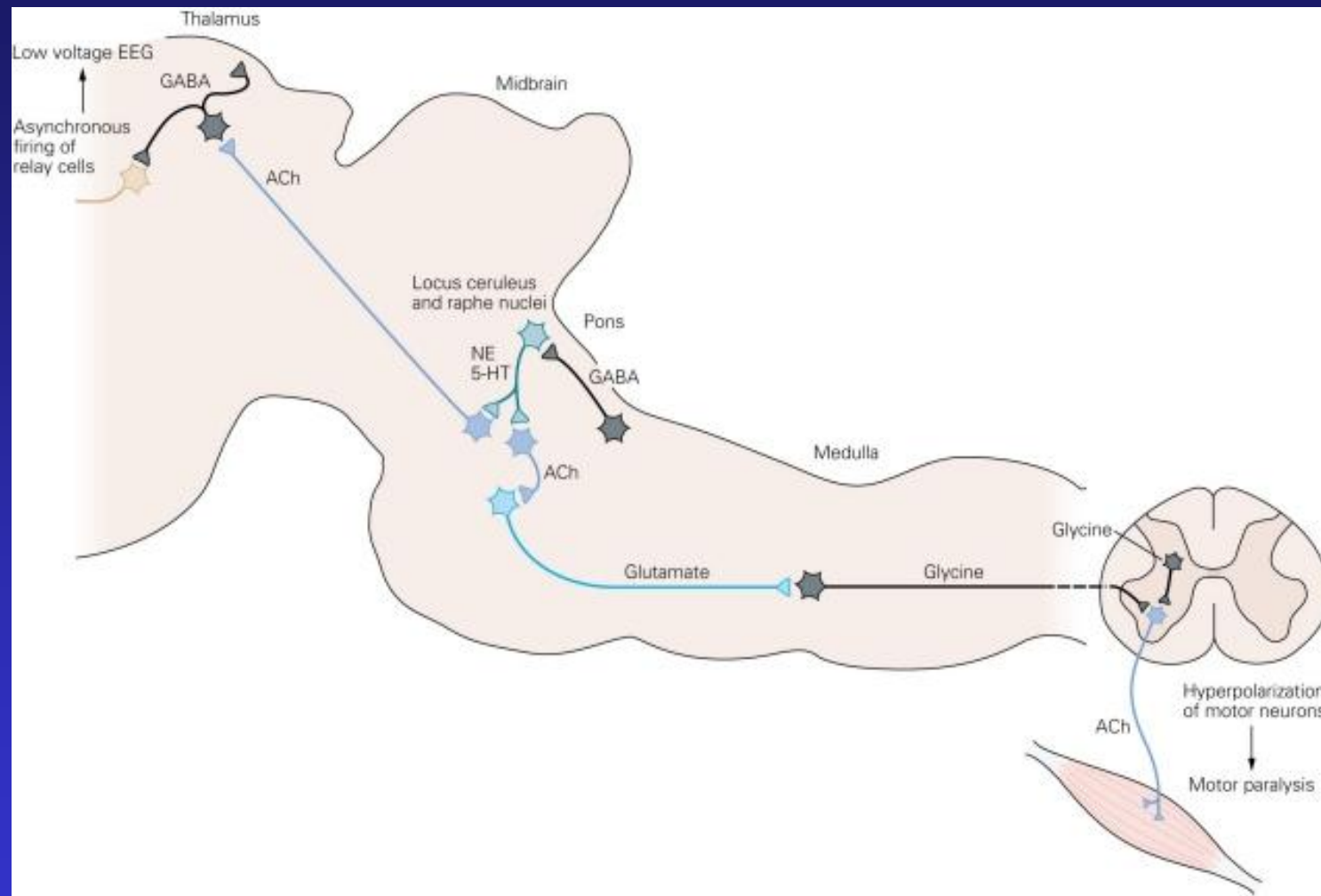
Avoid sleep fragmentation (OSA, alcohol)

Clonazepam 0.5-2.0 mg

REM Sleep Behavior Disorder

- Characterized by the intermittent loss of REM sleep atonia and by the appearance of elaborate motor activity associated with dream mentation
- Punching, kicking, leaping, and running from the bed during attempted dream enactment are frequent manifestations and usually correlate with the reported imagery
- Medical attention is often sought after injury has occurred to either the person or a bedpartner

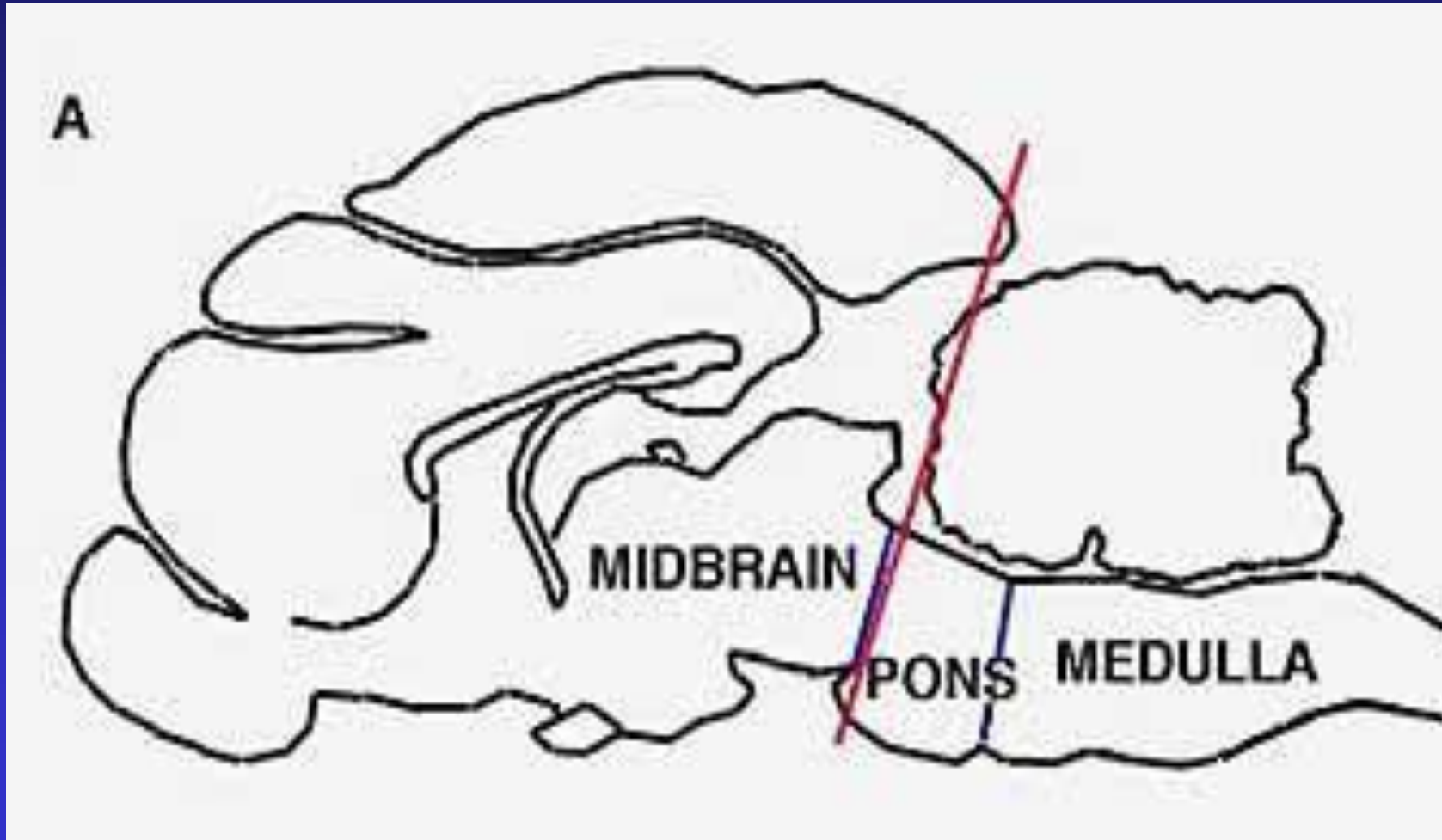
Cat Brainstem



REM Sleep Behavior Disorder

- An identical syndrome is seen in cats with experimentally induced bilateral peri-locus coeruleus lesions.
- Extensive neurologic evaluations in humans suffering from both the idiopathic and symptomatic forms have not identified specific lesions; however, findings in some patients suggest that diffuse lesions of the hemispheres, bilateral thalamic abnormalities, or primary brainstem lesions may result in the RBD

Jouvet & Cat Brain Transection



RBD

1

REM Sleep Behavior Disorder

- There may be a prodromal history of sleep talking, yelling, or limb jerking
- Idiopathic form more common in older men
- Secondary forms seen in neurodegenerative disorders such as Parkinson's, multiple sclerosis, olivopontocerebellar degeneration

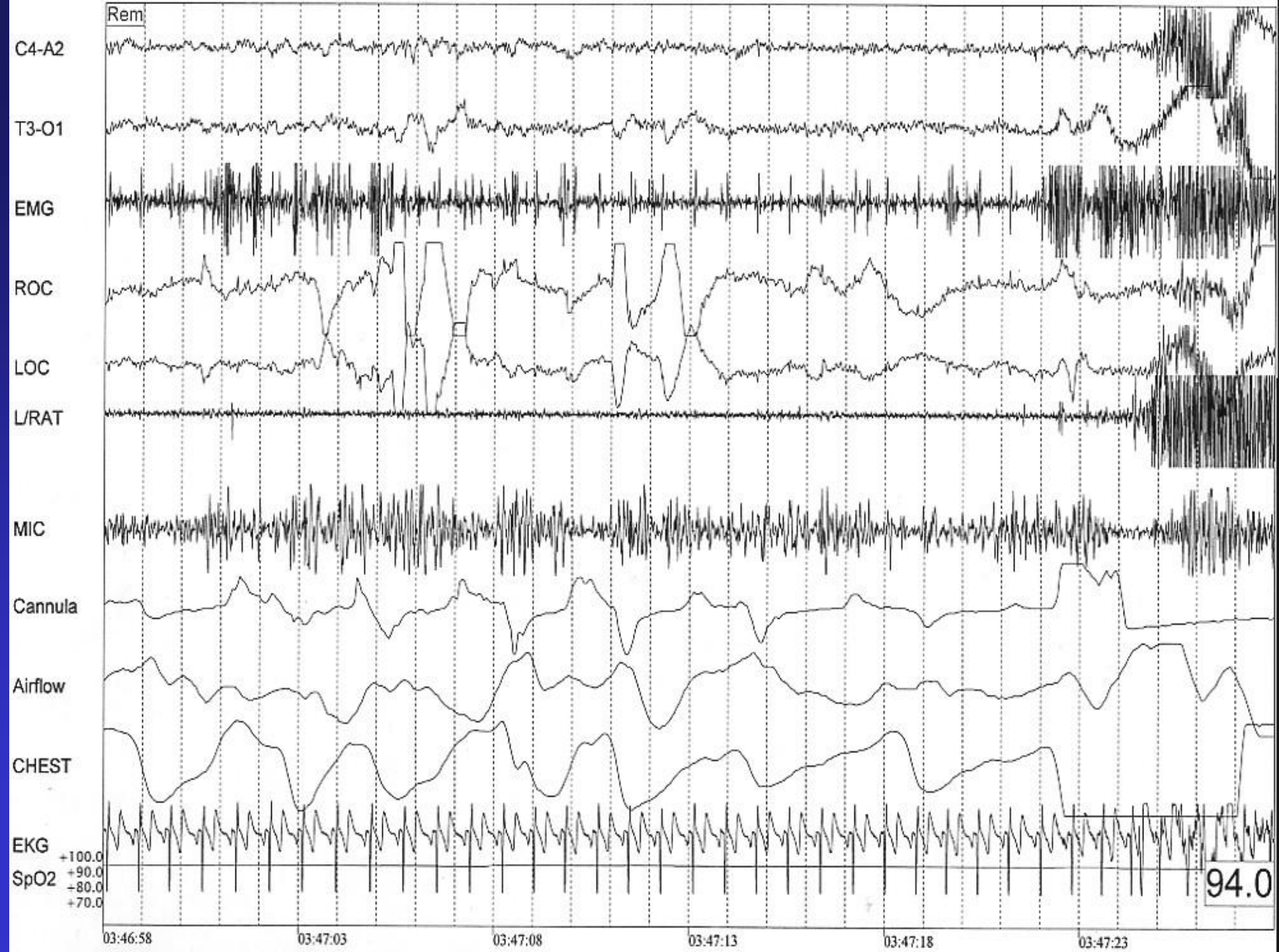
RBD

***REM SLEEP
BEHAVIOR
DISORDER***

RBD PSG Features

- REM displays persistent and possibly augmented muscle tone. Prominent and often prolonged periods of extremity activity are present. These motor phenomena may be highly integrated (repeated punching and kicking or more complex limb and trunk movements) and often are associated with emotionally charged utterances.
- In NREM, periodic movements involving the legs, and occasionally the arms, and periodic movements of all extremities is reported.
- There is frequently a pronounced increase in both the REM density and percentage of slow-wave sleep







RBD



REM Sleep Behavior Disorder Treatment

- Clonazepam 0.5-1.0 mg is effective ~ 90% with little tolerance or abuse
- Melatonin
- Safety:
 1. Remove dangerous objects/ weapons
 2. Cushions around the bed
 3. Mattress on the floor
 4. Windows covered



Sleep Forensics



In The Odyssey, the story of Elpenor, a companion of Ulysses, who got drunk before going to sleep. He was suddenly awakened from his sleep by a noise. In a confusional state, he forgot all about coming down by the main staircase, and he tumbled right off the roof and broke his neck (Homer, The Odyssey, Book 10)

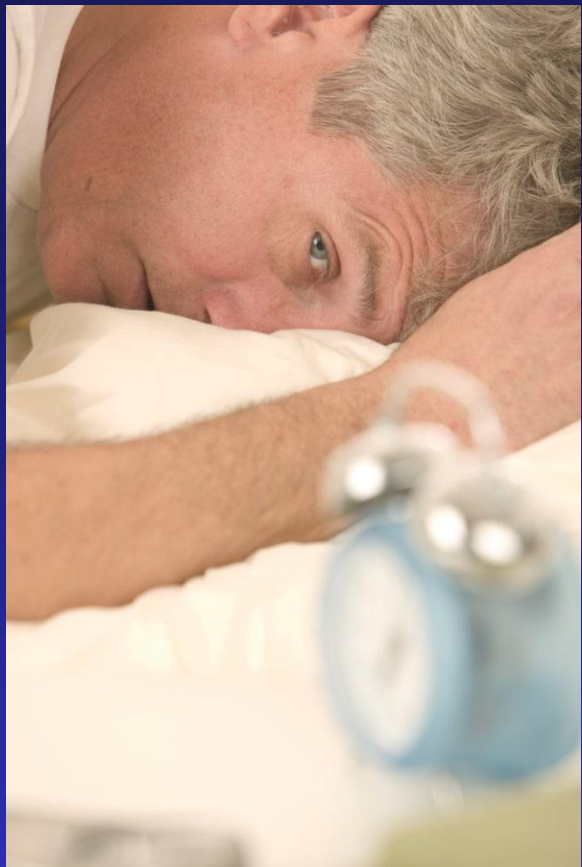




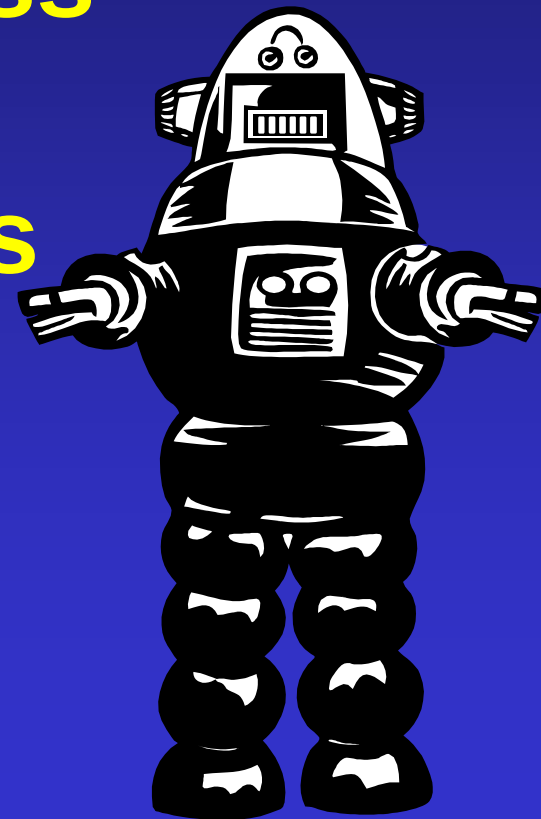
Conditions Associated with Sleep Related Violent Behavior

- Neurologic sleep disorders
- Disorders of arousal (confusional arousals [sleep drunkenness], sleepwalking, sleep terrors)*
- REM sleep behavior disorder*
- Nocturnal seizures* Automatic behavior
- Psychogenic sleep disorders-
- *Obstructive sleep apnea may mimic or trigger these parasomnias.

Anglo-American law defines criminal offenses whereby a person must be in a certain mental state, called the *mens rea* (guilty mind), necessary to have committed a crime. Persons possessed of *mens rea* cannot be convicted absent a corresponding criminal act, called the *actus reus* (guilty act). Traditionally, intention is found within *mens rea*, and the physical part of the offense resides within *actus reus*. Proof of both is essential to secure a conviction



Consciousness vs Automatisms



The first appearance of the “sleepwalking defense” in an American court of law came in *Massachusetts v Tirrell* in 1846. In this landmark case, Rufus Choate, a skilled orator and U.S. senator, successfully employed the “insanity of sleep” defense in the murder trial of Albert Tirrell. The evidence in that case proved that Tirrell brutally killed the victim with a razor, almost severing her head from her body, set the horrifically bloody crime scene ablaze, and then attempted to flee the country. Tirrell was acquitted.

Yellowlees in 1878

Young man with a history of sleep terrors who killed his 18-month-old son by smashing him against the wall during the night, taking him for a wild beast that was about to attack his family. He was acquitted. Special arrangement for sleeping created.

1992: Kenneth Parks

- A man drives 14 miles to the home of his in-laws. He man bludgeons his mother-in-law with a tire iron and stabs her with a knife and attempts to strangle his father-in-law. The woman dies, while her husband barely survives the attack. Parks cut the tendons in his own hands. Jury acquitted the defendant on the basis of automatism.
- Roger Broughton was the unchallenged sleep expert for the defense

Guidelines For Sleep Disorder & Violence

- There should be reason by history to suspect a bona fide sleep disorder. Similar episodes, with benign or morbid outcome, should have occurred previously.
- There has to be some degree of interaction with the environment. This behavior cannot be entirely passive in nature.
- The duration of the action is usually brief (seconds), although action of longer duration (minutes) does not necessarily exclude a sleep disorder or a sleep-related behavior. The action is usually abrupt, immediate, impulsive, and senseless—without apparent motivation. Although ostensibly purposeful, it is completely inappropriate to the total situation, out of (waking) character for the individual, and without evidence of premeditation.

Guidelines For Sleep Disorder & Violence

- The victim is someone who merely happened to be present, usually in close proximity, and who may have been the stimulus for the arousal. Sleepwalkers rarely, if ever, seek out victims.
- Immediately following return of consciousness, there is perplexity or horror, and there is no attempt to escape, conceal, or cover up the action. There is evidence of lack of awareness on the part of the sleepwalker during the event. There is usually some degree of amnesia for the event, but this amnesia need not be complete.

Guidelines For Sleep Disorder & Violence

- In the case of sleep terrors, sleepwalking, or sleep inertia, the act may occur on awakening (rarely immediately on falling asleep) and usually at least 1 hour after sleep onset. It occurs on attempts to awaken the subject. The action has been potentiated by sedative-hypnotics or by prior sleep deprivation. •
- **Lastly**, the violent behavior cannot be better explained by another mental disorder, medical condition, medication, or substance use. Ultimately, to attribute a violent behavior with criminal implications to parasomnia is a diagnosis of exclusion with the explicit understanding that other conditions are often more statistically likely.



Collaborative

Adversarial



Actual Cases

- Sleep conditions in solitary confinement
- Attempted murder of father- 16 year old
- Vandalism of neighbor's home- 10 yo boy
- Attempted rape - 19 year old college student
- Attempted murder of mother- 14 yo boy
- Hit and Run- off duty police officer
- and custody issues
- 7 cases of sexual molestation
- Airline pilot malpractice OSA
- Harbor pilot crashes into bridge
- Bus driver died while taking a nap
- Engineer sued over DSPS and car pool lane
- Prisoner sued over not getting narcolepsy meds
- Postal worker sued over chem spill
- Battery of Muni bus driver- homeless man
- 2 Family law cases involving sleep apnea

Parasomnia Takeaway

Conclusions

- Diverse group of clinical sleep disorders
- Some do not require active treatment, but injuries can occur
- Safety in the sleep environment needs to be stressed
- Violent behavior in sleep has forensic consequences
- Published forensic guidelines are available for legal proceedings

Questions?

